Twitter/X Data Extraction Pipeline

Overview

This document explains how to extract recent tweets related to each company using the Twitter/X API, process the data, and export it into a structured Excel dataset.

1. Obtain a Bearer Token (API Access)

1. Go to the Twitter Developer Platform.
2. Sign in with your Twitter/X account.
3. Apply for a Developer Account.
4. Create a Project and App.
5. Navigate to "Keys and Tokens".
6. Generate or copy your Bearer Token.

Note: Keep your Bearer Token secure and do not share it publicly.

2. Add Bearer Token to Code

Insert your token into the script:

```
BEARER_TOKEN = "YOUR_BEARER_TOKEN_HERE"
```

3. Define Search Query

Query used: Tesco delivery late OR Tesco -is:retweet

4. Send API Request

Send a GET request to the Twitter/X recent search endpoint with parameters such as query, max_results=100, and tweet fields.

5. Handle JSON Response

Extract tweet data from the JSON response using the "data" field.

6. Extract Tweet Data

Retrieve tweet ID, text, creation date, author ID, and engagement metrics (likes, retweets, replies, quotes).

7. Calculate Engagement

Engagement = likes + retweets + replies + quotes

8. Add Contextual Fields

Add fields such as Country (UK), Brand_Origin (Global), Retailer (Amazon), and Source_Platform (Twitter/X).

9. Structure Data

Convert extracted rows into a pandas DataFrame.

10. Export Dataset

Save the dataset as x_scraped_dataset.xlsx.

Output

The script generates an Excel file containing structured tweet data ready for analysis.

The same code was followed by changing the company name to obtain data for different companies and changing if they are global or local companies.

CODE:

```
import requests
import pandas as pd
from datetime import datetime

BEARER_TOKEN = "YOUR_BEARER_TOKEN_HERE"

def search_tweets(query, max_results=100):  # Changed to 100
    url = "https://api.twitter.com/2/tweets/search/recent"
    headers = {
        "Authorization": f"Bearer {BEARER_TOKEN}"
    }
    params = {
        "query": query,
        "max_results": max_results,
        "tweet.fields": "created_at,public_metrics,lang,source",
        "expansions": "author_id",
        "user.fields": "id,name,username"
    }
    response = requests.get(url, headers=headers, params=params)
    return response.json()

def build_dataset(json_response):
    rows = []
```

```
    tweets = json_response.get("data", [])
    for t in tweets:
        metrics = t["public_metrics"]
        row = {
            "Country": "UK",  # Your contextual variable
            "Brand_Origin": "Global",
            "Retailer": "IKEA",  # Changed to IKEA
            "Source_Platform": "Twitter/X",
            "Post_ID": t["id"],
            "Date": t["created_at"][:10],
            "Author_ID_Anon": f"user_{t['author_id']}",
            "Engagement_Count": metrics["retweet_count"] + metrics["like_count"] +
metrics["reply_count"] + metrics["quote_count"],
            "Post_Text": t["text"],
            "Sentiment_VADER": ""  # to be added later
        }
        rows.append(row)
    return rows


# ---- RUN SEARCH ----
query = "Amazon delivery late OR Amazon Prime -is:retweet"
json_data = search_tweets(query)
rows = build_dataset(json_data)
df = pd.DataFrame(rows)
output_path = "x_scraped_dataset.xlsx"
df.to_excel(output_path, index=False)
```

```
print(f"Dataset saved to {output_path}")
```